

## \* Digital - To - Digital Conversion

The conversion of digital data into digital signal involves three techniques - line coding, block coding and scrambling.

### \* Line Coding

- Line coding is the process of converting digital data to digital signals.
- Line coding converts a sequence of bits to a digital signal.
- At the sender, digital data are encoded into a digital signal; at the receiver, the digital data are recreated by decoding the digital signal.

### Characteristics:

- Data Element: A data element is the smallest entity that can represent a piece of information (bit).
- Signal Element: A signal element is the shortest unit (time wise) of a digital signal.
- Data elements are what we need to send, signal elements are what we can send.
- Data Rate: No of data elements that we can transfer per second.  
It is denoted by  $N$  and has unit bps.
- Signal Rate: Also called as baudrate.  
It is the number of signal elements moved per second.  
It is denoted by  $S$  and unit is baud.

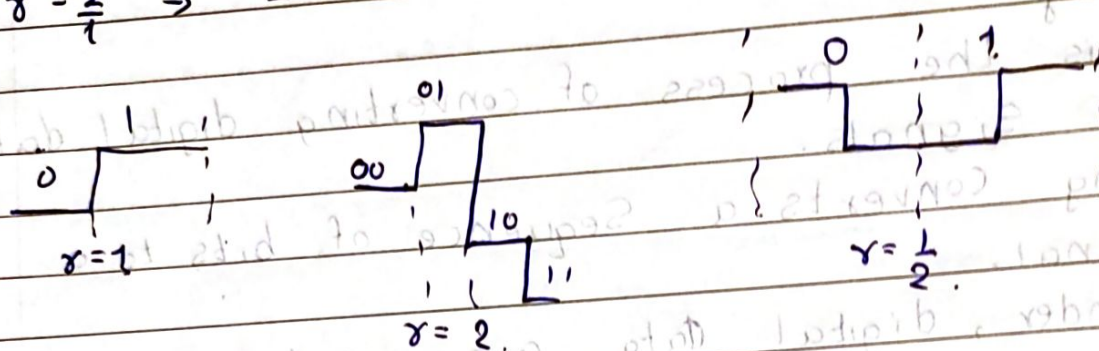
We define a ratio  $r$  which is the number of data elements carried by each signal element.

$$r = \frac{\text{No of data element}}{\text{No of Signal element}}$$

$$\text{Bitrate} = \text{Baudrate} \times \text{Bits per Symbol}$$

$$\text{Bitrate} = \text{Baudrate} \text{ if } r = 1$$

$$r = \frac{2}{1} \Rightarrow 2 \text{ data elements per Signal element}$$



$$S = C \times N \times \frac{1}{r} \text{ baud}$$

$S = \text{baudrate}$ ,  $N = \text{data rate}$ ,  $C = \text{Case factor } (\frac{1}{2})$

$r = \text{ratio}$

### \* Characteristics

i) **Baseline Wandering**: In decoding a digital signal, the receiver calculates a running average of the received signal power. This average is called the baseline.

The incoming signal power is evaluated against this baseline to determine the value of the data element.

A long string of 0s or 1s can cause a drift in the baseline (baseline wandering) and make it difficult for the receiver to decode correctly.

A good line coding scheme needs  $\phi$  to prevent baseline wandering.

Baseline wandering occurs when the receiver loses track of the average voltage level (baseline) due to long sequences of 0s or 1s.

- Long Sequence of all 1s or all 0s cause the signal's average to drift.

ii) DC Components: When the voltage level in a digital signal is constant for a while, the spectrum creates very low frequencies (results of Fourier analysis).

- These frequencies, around zero, called DC components, present problems for a system that cannot pass low frequencies or a system that uses electrical coupling. (via transformer).

• Ex. a telephone line cannot pass frequency below 200 Hz.

• A DC component is the non zero average voltage in a signal. It is problematic because:

- Some transmission media (ex. transformer) cannot pass DC signals.

• DC bias can cause distortion.

⇒ • Ideally a transmission signal should have an average voltage of zero to ensure efficient transmission.

- If a signal has DC component, some devices will filter it out ex. transformer will filter it out, causing distortion.

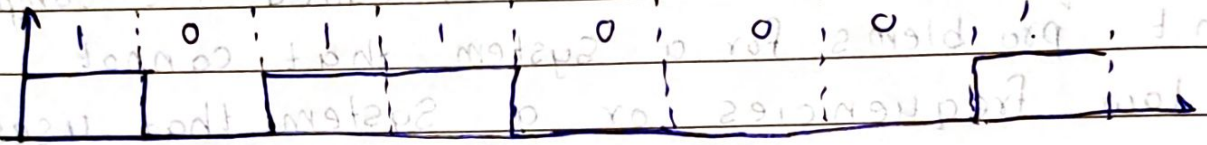
• DC components consume power unnecessarily, reducing efficiency.

- They may cause heating.
- It may affect clock synchronization in communication systems. (There should be minimum DC components).

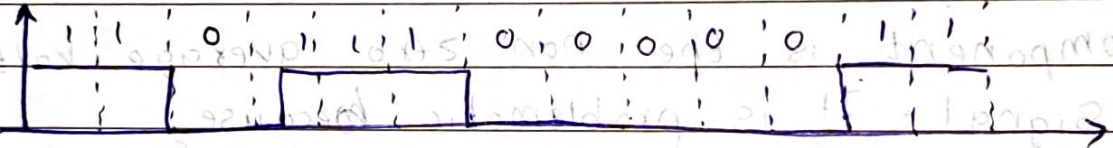
iii) Self Synchronization: To correctly interpret the signals received from the sender, the receiver's bit intervals must correspond exactly to the sender's bit intervals.

If the receiver clock is faster or slower, the bit intervals are not matched.  
ex. if sender is sending 1 bps and receiver is receiving 2bps causing wrong signals being received and receiver might misinterpret the signals.

ex.



a. Sent (10110001)



b. received (110111000011)

A Self-Synchronizing digital signal includes timing information in the data being transmitted.

This can be achieved if there are transitions in the signal that alert the receiver to beginning, middle or end of the pulse.

- If the receiver's clock is out of synchronization, these points can reset the clock.

iv > Built-in Error Detection: It is desirable to have a builtin error detecting capability in the generated code to detect some or all the errors that occurred during transmission.

- Some encoding schemes have this capability.
- It allows the receiver to identify corrupted data.

v) Immunity to Noise and Interference: Another desirable code characteristic is a code that is immune to noise and other interferences. Noise resistance ensures that minor signal distortions do not cause bit errors.

vi) Complexity: A complex Scheme is more costly to implement than a simple one. For ex. a Scheme that uses four signal levels is more difficult to interpret than one that uses only two levels.

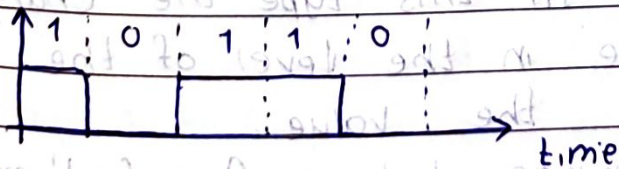
- Some encoding Schemes are easier to implement, while others require complex hardware.

### \* Line Coding Schemes:

\* Unipolar Scheme: In a unipolar Scheme, all the signal levels are on one side of the time axis, either above or below.

i) Unipolar NRZ (Non-Return to Zero)

It is a unipolar line coding Scheme in which positive voltage defines bit 1 and zero voltage defines bit 0. Signal does not return to zero at the middle of the bit thus it is called NRZ.



- But this Scheme uses more power as compared to polar Schemes to send one bit per unit line resistance.

- If a continuous set of zero's or one's there will be self-synchronization and base line wandering problem.

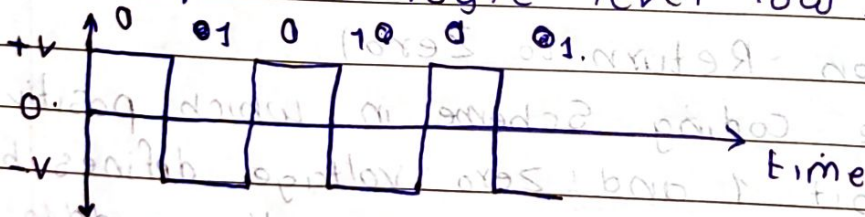
- This Scheme is very costly compared to polar.
- Normalized power (power needed to send 1 bit per unit line resistance) is double that for polar NRZ.

### \* Polar Schemes:

In polar schemes, the voltages are on the both sides of the time axis.  
ex. the voltage for 0 can be positive and voltage for 1 can be negative.

1) Polar NRZ : In polar NRZ encoding, we use two levels of voltage amplitude.  
~~usually~~ We can have two versions of polar NRZ.

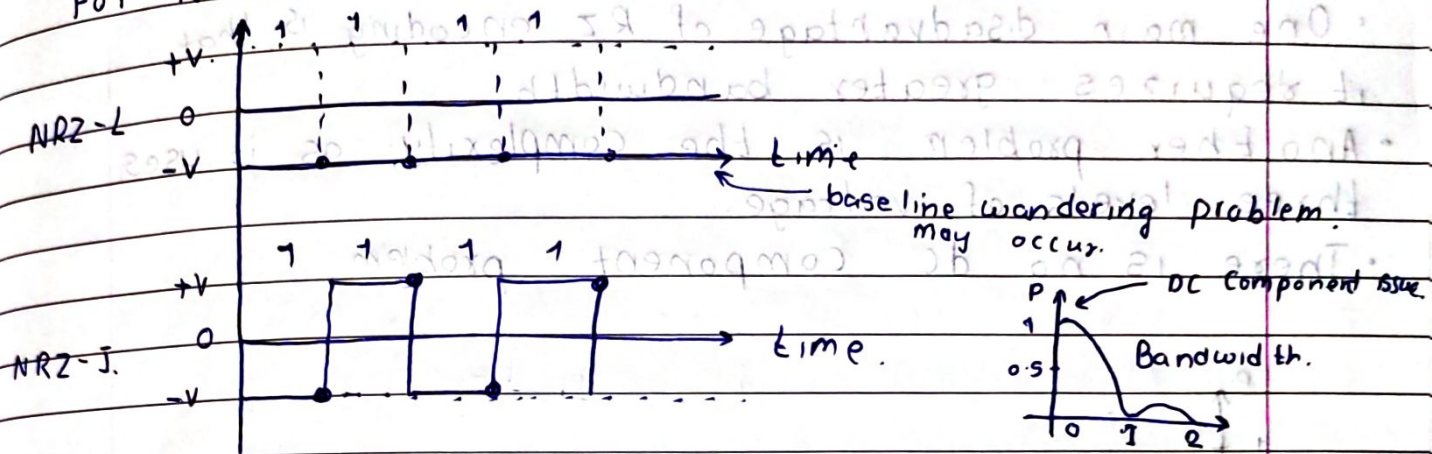
1) Polar NRZ - L (Level) : In this the level of the voltage determines the value of the bit.  
ex. binary 1 maps to logic-high and binary 0 maps to logic-level low.



2) NRZ - I (Invert) : In this type the change or lack of change in the level of the voltage determines the value.  
• If there is no change bit is 0 ; if there is change, bit is 1.



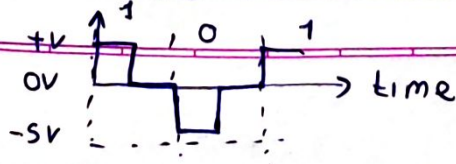
- Baseline wandering is a problem for both of them, but in NRZ-L it is twice as bad as compared to NRZ-I.
- This is because of transitions at the boundary for NRZ-I.

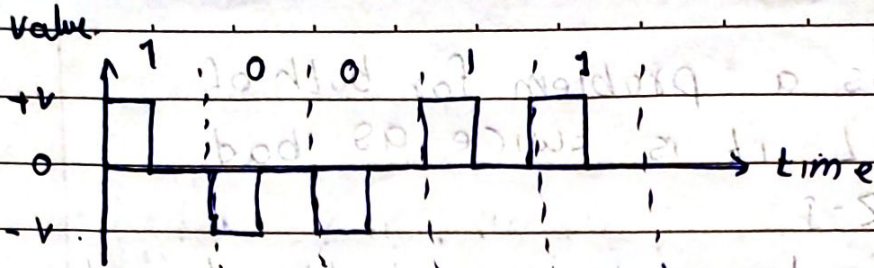


- NRZ-I and NRZ-L both have DC component problem.
- Similarly Self-Synchronization problem is similar in both for long sequence of 0's but for long sequence of 1's it's more severe in NRZ-L.
- Both have signal rate of  $N/2$  Baud.

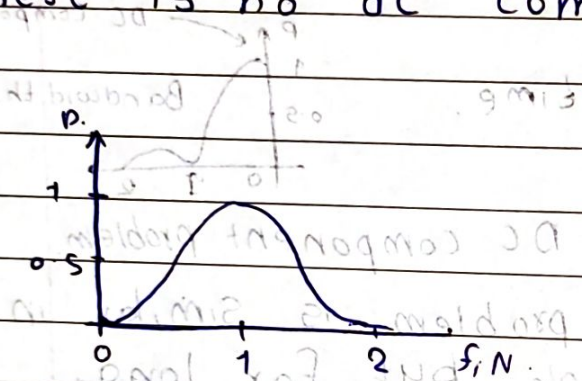
### 3) Return to Zero (RZ)

- The main problem with NRZ encoding occurs when the sender and receiver clocks are not synchronized.
- The receiver does not know when one bit has ended and the next bit is starting.
- To resolve this RZ scheme is used, which uses three values: positive, negative, and zero.
- In RZ, the signal changes not between bits but during the bit.
- ex. to show 0 signal goes to level 0 in middle of bit and stays there till start of next bit and then moves ~~middle~~ to positive or negative





- One main disadvantage of RZ encoding is that it requires greater bandwidth.
- Another problem is the complexity as it uses three levels of voltage.
- There is no DC component problem.



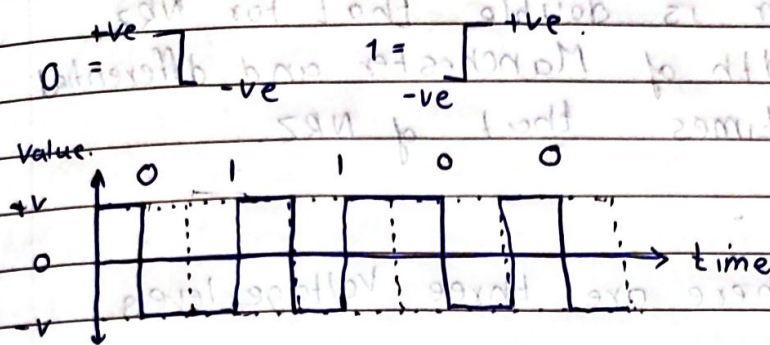
In NRZ the power density is very high around frequency close to zero. This means that there are DC components that carry a high level of energy.

This is not the case in RZ.

#### 4) Biphasic Manchester

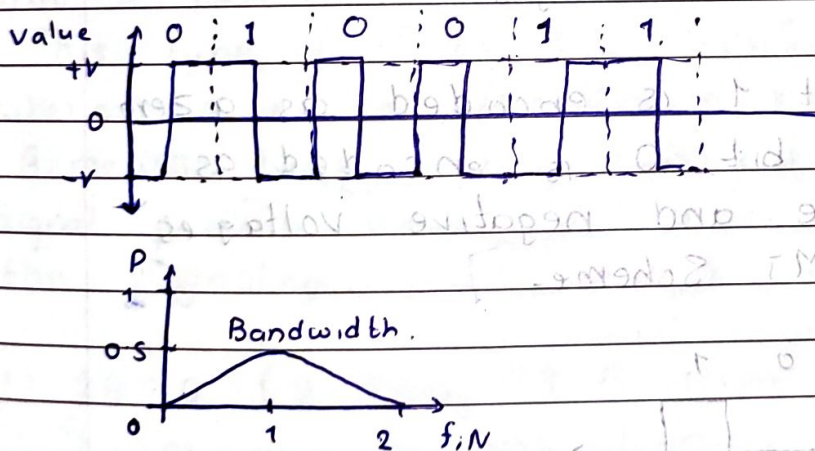
- Manchester encoding is somewhat combination of the RZ (transition at the middle of the bit) and NRZ-L Schemes.
- In Manchester encoding, the duration of the bit is divided into two halves.
- The voltage remains at one level during the first half and moves to other level in the second half.

- The transition at the middle of the bit provides Synchronization.



### 5) Biphase Differential Manchester.

- Differential Manchester combines ideas of RZ and NRZ-I. There is always transition at the middle of the bit.
- If next bit is 0 there is an inversion and if next bit is 1, no transition (inversion).



- The Manchester scheme overcomes several problems associated with NRZ-L and differential Manchester overcomes several problems associated with NRZ-I.
- Firstly, there is no baseline wandering.
- There is no DC component because each bit has a positive and negative voltage contribution.
- Differential Manchester is used in LAN.

- The only drawback is signal rate.
- The signal rate for Manchester and differential Manchester is double that for NRZ.
- The minimum bandwidth of Manchester and differential Manchester is 2 times that of NRZ.

### \* Bipolar Schemes:

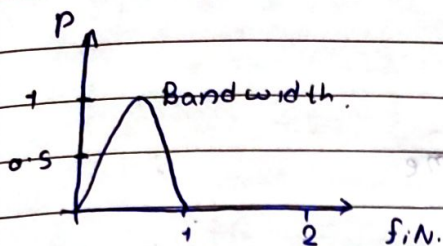
- In this scheme there are three voltage levels positive, and zero.
- The voltage level for one data element is at zero, while the voltage level for the other element alternates between positive and negative.

1) Alternate Mark Inversion (AMI): A neutral zero voltage <sup>represent binary 0</sup> and the bit 1 is encoded as alternating positive and negative voltages.

2) Pseudoternary - Bit 1 is encoded as a zero voltage and the bit 0 is encoded as alternating positive and negative voltages i.e opposite of AMI Scheme.



- The bipolar scheme is an alternative to NRZ. This scheme has the same signal rate as NRZ but there is no DC component.



- AM is commonly used for long-distance communication but it has a synchronization problem when a long sequence of 0s is present.

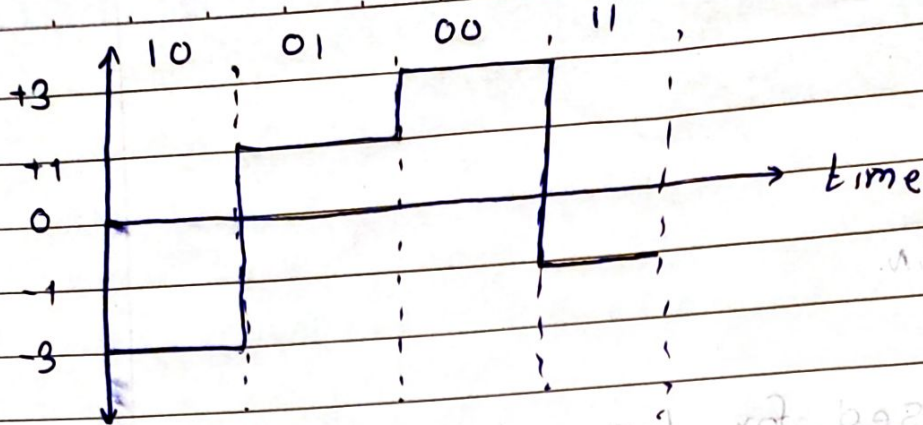
### \* Multilevel Schemes :

- The desire to increase the data speed or decrease the required bandwidth has resulted in creation of many schemes.
- The goal is to increase the number of bits per baud by encoding a pattern of  $m$  data elements into a pattern of  $n$  signal elements.
- This type of coding is classified as  $mBnL$ , where  $m$  is the length of the binary pattern,  $B$  means binary data,  $n$  is the length of the signal pattern, and  $L$  is the number of levels in the signaling.

#### 1) 2B1Q (2 Binary 1 Quaternary)

- It uses data patterns of size 2 and encodes the 2-bit patterns as one signal element belonging to a four-level signal.
- In this type  $m=2$ ,  $n=1$ ,  $L=4$  (Quaternary).
- The average signal rate of 2B1Q is  $S = N/4$ . This means that using 2B1Q, we can send data 2 times faster than NRZ-L.
- However it uses 4 voltage levels, so receiver has to discern four different thresholds.

• 2B1Q is used in DSL (Digital Subscriber Line) to provide high-speed connection to internet.



Suppose

|    |   |     |
|----|---|-----|
| 00 | = | +3V |
| 01 | = | +1V |
| 10 | = | -3V |
| 11 | = | -1V |

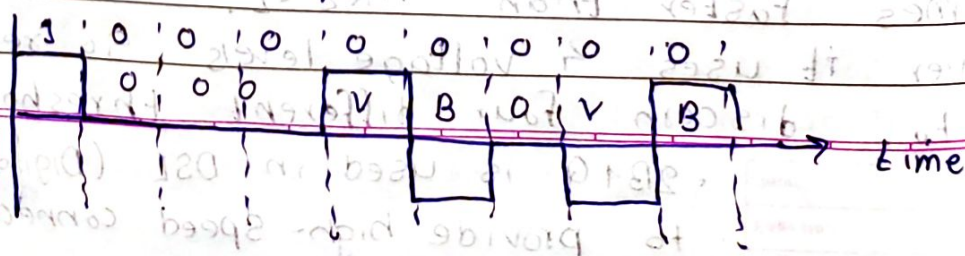
### \* Scrambling :

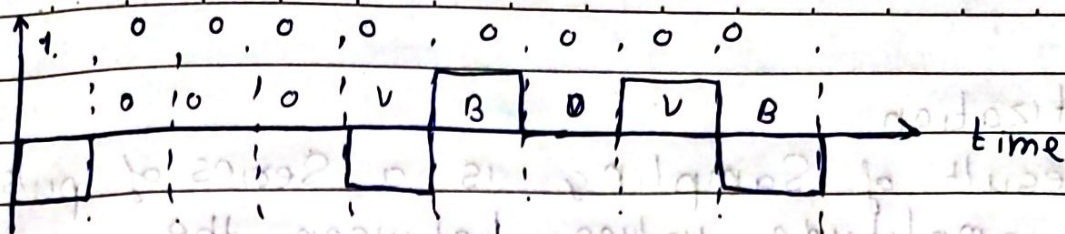
Scrambling is modifying a data sequence to prevent long runs of identical bits (eg. consecutive zeros) which can cause synchronization issues.

#### 1) B8ZS (Bipolar 8-Zero Substitution)

- In this technique, eight consecutive zero level voltages are replaced by the sequence 000VB0VB.
- The V in sequence denotes Violation.
- V is non zero voltage that breaks an AMI rule of encoding.
- The B in the sequence denotes bipolar, which means a non zero level according with AMI rules.
- It is a modified AMI Scheme.
- There are two cases.

a. Previous level is positive.

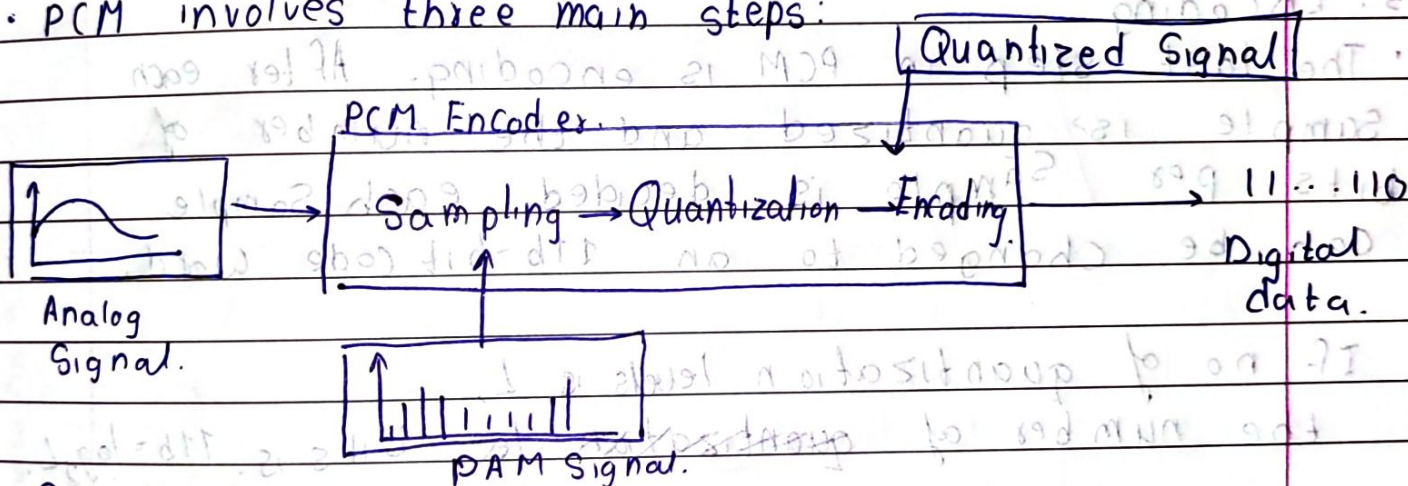




b. Previous level is negative

### \* Pulse Code Modulation:

- The most common technique to change an analog signal to digital data (digitization) is called pulse code modulation (PCM).
- It is widely used in digital audio, telecommunications, and signal processing.
- PCM involves three main steps:



#### 1. Sampling:

- The first step in PCM is Sampling. The analog signal is sampled every  $T_s$  s, where  $T_s$  is the sample interval or period.
- The inverse of the sampling interval is called the sampling rate or sampling frequency and is denoted by  $f_s$ .
- The sampling rate must be at least twice the highest frequency of the signal (According to Nyquist theorem).

## 2. Quantization

- The result of Sampling is a Series of pulses with amplitude values between the maximum and minimum amplitudes of the Signal
- Each Sample value is approximated to nearest value from a finite set of discrete levels.
- The difference between the actual Sample value and the quantized value is called quantization error.

## 3. Encoding

- The last step in PCM is encoding. After each Sample is quantized and the number of bits per Sample is decided, each Sample can be changed to an  $n$ -bit code word.
- If no of quantization levels is  $L$ , the number of ~~quantization~~ bits is  $n = \log_2 L$ .
- As we increases the levels the quantization error reduces.

Q. Find the encoded digital code for-

1.  $-6.3V$ , 2.  $-15.9V$ , 3.  $-9.7V$ , 4.  $7.8V$ .  
5.  $12.3V$ , 6.  $14.10V$ .

| Values | Normalized PMV | Normalized |
|--------|----------------|------------|
|--------|----------------|------------|

| Value  | Normalized PMV | Normalized Quantized | Quantization Error | Quantization Code | Encoded Word |
|--------|----------------|----------------------|--------------------|-------------------|--------------|
| -6.3   | -1.26          | -1.5                 | -0.24              | 2                 | 010          |
| -15.9  | -3.18          | -3.5                 | -0.32              | 0                 | 1000         |
| -9.7   | -1.94          | -1.5                 | -0.44              | 2                 | 010          |
| 7.8    | 1.56           | 1.5                  | 0.06               | 5                 | 101          |
| 12.3   | 2.26           | 2.5                  | 0.24               | 6                 | 110          |
| 14.366 | 2.92           | 2.5                  | 0.42               | 6                 | 110          |

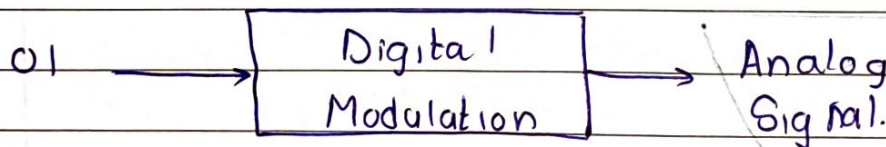
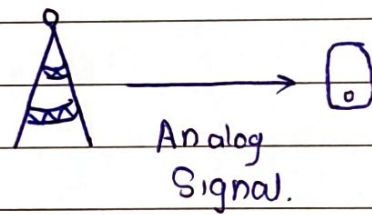
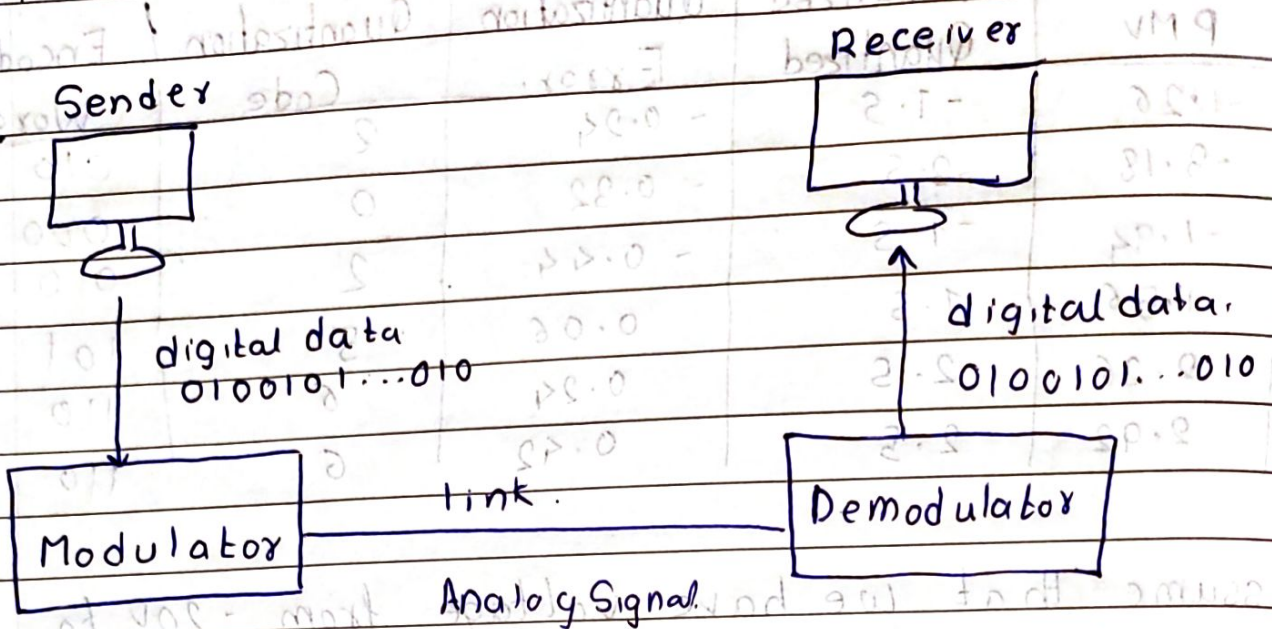
Assume that we have voltage from  $-20V$  to  $+20V$ . We are using 8 levels.

$$\Delta = \frac{V_{max} - V_{min}}{\text{No of Levels}} = \frac{20 - (-20)}{8} = 5V$$

$$\Delta = 5V$$



- \* Digital to Analog Conversion
- Digital to Analog Conversion (DAC) is the process of converting digital signals (discrete values) into analog signals (continuous waveforms).
- This is crucial in communication systems where digital data must be transmitted over analog channels, such as radio waves or telephone lines.



Digital modulation is a specific type of DAC where digital signals modulate a carrier wave (analog signal) for transmission.

1. **Amplitude Shift Keying (ASK)** : It is a digital modulation technique where the amplitude of a carrier wave is varied according to the digital data being transmitted while keeping the frequency and phase constant.

- RFID (Radio Frequency Identification)
- Optical fiber communication

Also called On Off keying. OOK.

- Bit 1  $\rightarrow$  High Amplitude.
- Bit 0  $\rightarrow$  Low (or zero) amplitude.

#### \* Advantages.

- Simple to implement.
- Requires less bandwidth.

#### \* Disadvantages.

- Highly Susceptible to noise.
- Signal strength variation (attenuation) can cause data loss.

if  $i/p \leftarrow 0$

$$S(t) = A_1 \sin 2\pi f t$$

else

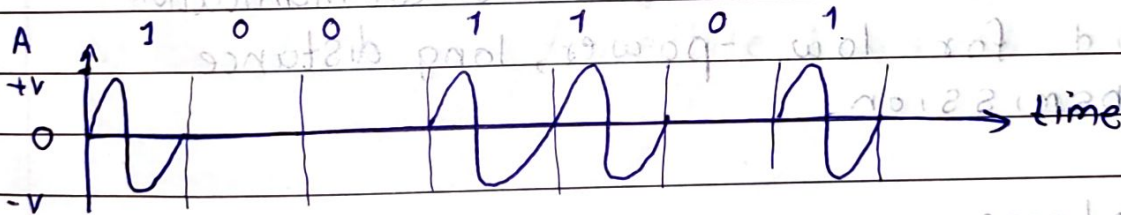
$$S(t) = A_2 \sin 2\pi f t$$

if  $S(t) = A_1 \sin 2\pi f t$  then

bit  $\leftarrow 1$

else

bit  $\leftarrow 0$



#### \* Frequency Shift keying (FSK):

Frequency shift keying (FSK) is a digital modulation technique where the frequency of a carrier wave is varied according to the digital data being transmitted while keeping the amplitude and phase constant.

Bluetooth & wireless communication

• Low power IoT devices.

• Radio teletype.

Bit 1  $\rightarrow$  High frequency  
 Bit 0  $\rightarrow$  Lower frequency.

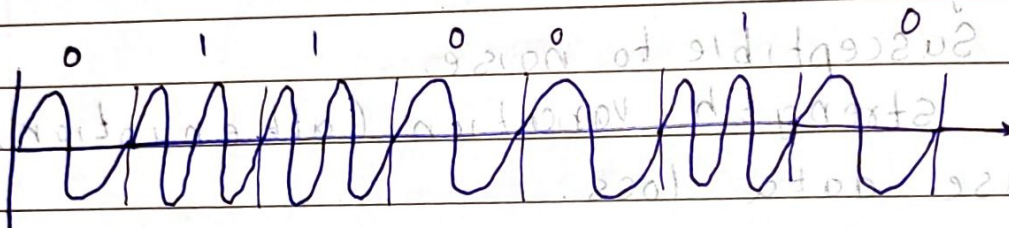
Scan i/p

if i/p  $\leftarrow$  1

$$S(t) = A \sin 2\pi f_1 t$$

else i/p  $\leftarrow$  0

$$S(t) = A \sin 2\pi f_2 t$$



Assume  $f_1 = 2f_2$

Bit 1  $\rightarrow f_1 = 2f_2$   
 Bit 2  $\rightarrow f_2$

#### \* Advantages $\rightarrow$

- More noise resistant
- Used in wireless & radio communication
- Good for low-power, long distance transmission.

#### \* Disadvantages

- Require more bandwidth than ASK.
- Hardware complexity is higher than ASK.

#### \* Binary Phase Shift Keying (BPSK):

It is a digital modulation technique where the phase of a carrier wave is shifted between two distinct values ( $0^\circ$  and  $180^\circ$ ) to represent binary data (0 and 1).

Bit 0  $\rightarrow$  Phase  $\phi_1$

Bit 1  $\rightarrow$  Phase  $\phi_2$

Scan i/p.

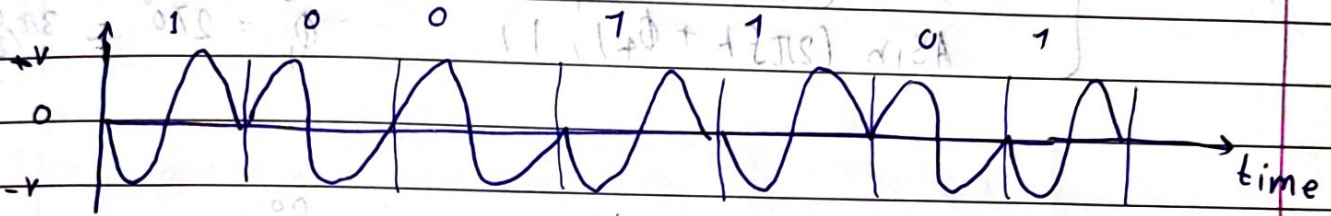
if i/p  $\leftarrow$  0

$$S(t) = A \sin(2\pi f t + \phi_1)$$

else if i/p  $\leftarrow$  1

$$S(t) = A \sin(2\pi f t + \phi_2)$$

end if



#### \* Advantages

- Simple and robust against noise.
- Efficient power usage.
- Widely used in Satellite & wireless communication.

#### \* Disadvantages

- Slow data rate.
- Complex receiver design.

$$\text{Baudrate} = \text{Bitrate} \quad (\text{ASK, FSK, BPSK})$$

#### \* Quadrature Phase Shift Keying (QPSK):

Quadrature phase shift keying is a digital modulation technique where the phase of the carrier wave is varied among four

distinct values. ( $0^\circ, 90^\circ, 180^\circ, 270^\circ$ ) to represent two bits per Symbol.

4G LTE, 5G, Wi-Fi 802.11

Satellite Communications (DVB-S)

Cable TV, DSL, Internet

$00 \rightarrow 0^\circ$

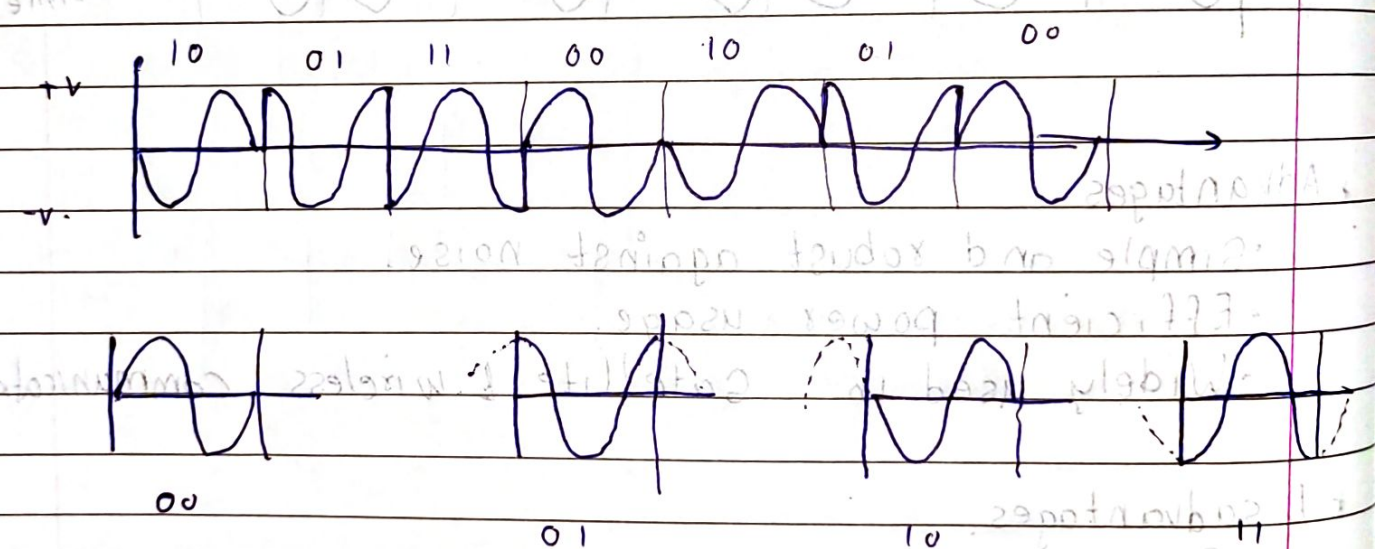
$01 \rightarrow 90^\circ$

$10 \rightarrow 180^\circ$

$11 \rightarrow 270^\circ$

$$S(t) = \begin{cases} A \sin(2\pi f t + \phi_1), & 00 \\ A \sin(2\pi f t + \phi_2), & 01 \\ A \sin(2\pi f t + \phi_3), & 10 \\ A \sin(2\pi f t + \phi_4), & 11 \end{cases}$$

$\phi_1 = 0^\circ = 0$   
 $\phi_2 = 90^\circ = \pi/2$   
 $\phi_3 = 180^\circ = \pi$   
 $\phi_4 = 270^\circ = 3\pi/2$



### \* Advantages of QPSK

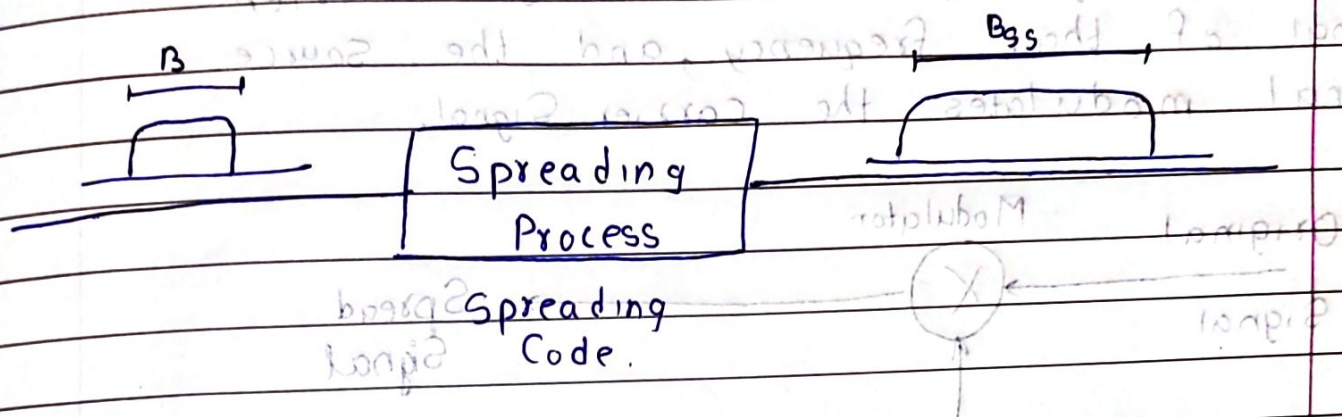
- Higher data rate than BPSK (transmits 2 bit per Symbol)
- More bandwidth efficient than BPSK
- Same noise resistance as BPSK

### \* Disadvantages

- More Complex receiver design.
- More Susceptible to phase noise than BPSK.

## \* Spread Spectrum

- Spread Spectrum is a technique used in wireless communication to spread a signal over a wider bandwidth than necessary.
- It improves security, resistance to interference and efficient frequency utilization.



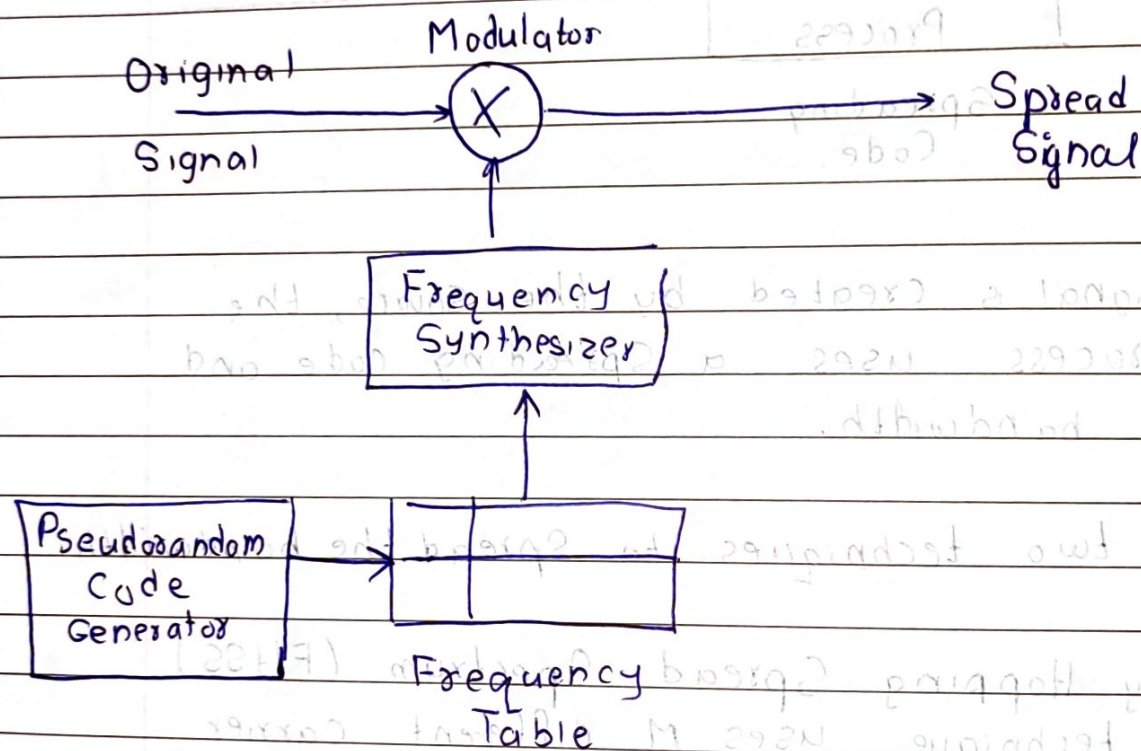
- After the signal is created by the source, the spreading process uses a spreading code and spreads the bandwidth.

There are two techniques to spread the bandwidth:

### 1. Frequency Hopping Spread Spectrum (FHSS)

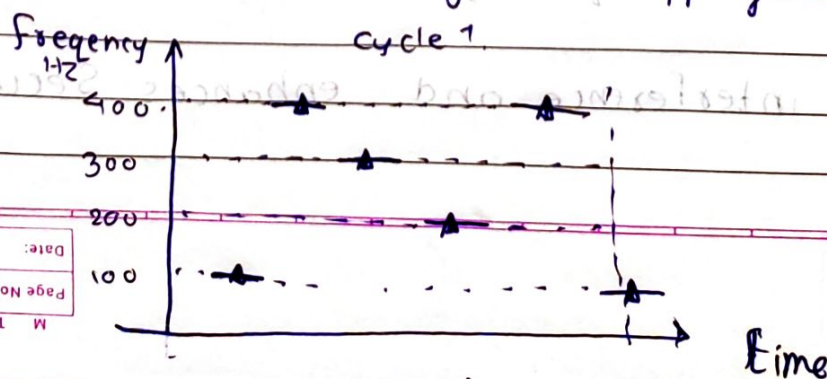
- The FHSS technique uses  $M$  different carrier frequencies that are modulated by the source signal.
- At one moment, the signal modulates one carrier frequency, at the next moment, the signal modulates another carrier frequency.
- The transmitted signal rapidly switches (hops) between different frequency channels in a predefined sequence.
- This reduces interference and enhances security.

- A pseudorandom code generator called pseudorandom noise (PN), creates a 1k-6H pattern for every hopping period  $T_h$ .
- The frequency table uses the pattern to find the frequency to be used for this hopping period and passes it to the frequency synthesizer.
- The frequency synthesizer creates a carrier signal of that frequency, and the source signal modulates the carrier signal.



### \* key parameters of FHSS

- Hopping rate - The speed at which freq change
- Hopping Sequence - A pseudorandom sequence known to both sender & receiver
- Dwell Time - Time spent on a single freq before hopping



### \* Advantages:

- Resistance to Narrowband Interference.
- Hard to intercept without knowing hopping Sequence.
- Efficient use of Spectrum

### 2. Direct Sequence Spread Spectrum (DSSS)

- In DSSS, we replace each data bit with  $n$  bits using a Spreading code.
- In other words, each bit is assigned a code of  $n$  bits called chips, where chip rate is  $n$  times that of the data bit.
- In this the data signal is multiplied by a high-rate pseudorandom bit sequence (Chipping Code), (e.g. Barker Code).

